

Toll_Plaza	Plaza_ID -> Plaza_Name, Location, Highway_Name, Number_of_Lanes, Contact_Number
Toll_Booth	Booth_ID -> Plaza_ID, Booth_Number, Status Plaza_ID, Booth_Number -> Booth_ID (alternate key if Booth_Number unique per plaza)
Staff	Staff_ID -> Staff_Name, Role, Salary, Booth_ID, Shift_Start, Shift_End
Staff_Plaza	Staff_ID, Plaza_ID -> (no non-key attributes)
Toll_Rate	Rate_ID -> Plaza_ID, Vehicle_Type, Toll_Amount Plaza_ID, Vehicle_Type -> Rate_ID, Toll_Amount

Vehicle	Vehicle_ID -> Vehicle_Number, Vehicle_Type, Owner_Name, Owner_Contact_Number
Transaction	Transaction_ID -> Vehicle_ID, Booth_ID, Transaction_Time, Amount_Paid, Payment_Mode
Monthly_Pass	Pass_ID -> Vehicle_ID, Plaza_ID, Expiry_Date, Pass_Fee
Fastag_Account	Fastag_ID -> Vehicle_ID, Balance, Bank_Name
Violation	Violation_ID -> Vehicle_ID, Plaza_ID, Violation_Type, Violation_Date, Penalty_Amount
Transaction_Pass	Transaction_ID, Pass_ID -> (no non-key attributes)
Transaction_Fastag	Transaction_ID, Fastag_ID -> (no non-key attributes)
Transaction_Rate	Transaction_ID, Rate_ID -> (no non-key attributes)

3. BCNF Check:

- **Toll_Plaza**

Candidate Keys	{ Plaza_ID }
BCNF Check	For every FD $X \rightarrow Y$: $X = \text{Plaza_ID}$ which is a superkey. No partial or transitive dependencies exist.
Verdict	IN BCNF. No decomposition needed.

- **Toll_Booth**

Candidate Keys	{ Booth_ID } alternate: { Plaza_ID, Booth_Number } (if Booth_Number is unique per plaza)
-----------------------	--

BCNF Check	The only non-trivial FD has Booth_ID as the determinant, which is a superkey. The alternate key FD (Plaza_ID, Booth_Number -> Booth_ID) also has a superkey on the left.
Verdict	IN BCNF. No decomposition needed.

- **Staff**

Candidate Keys	{ Staff_ID }
BCNF Check	All attributes are fully functionally dependent on Staff_ID. Booth_ID is a FK (not a determinant of other non-key attributes within this relation). No transitive FDs within the relation itself.
Verdict	IN BCNF. No decomposition needed.

- **Staff_Plaza**

Candidate Keys	{ Staff_ID, Plaza_ID }
BCNF Check	A relation with only key attributes is trivially in BCNF.
Verdict	IN BCNF. No decomposition needed.

- **Toll_Rate**

Candidate Keys	{ Rate_ID } alternate: { Plaza_ID, Vehicle_Type }
BCNF Check	FD1: Rate_ID is a superkey. FD2: {Plaza_ID, Vehicle_Type} is an alternate candidate key, hence a superkey. Both FDs satisfy BCNF.
Verdict	IN BCNF. No decomposition needed.

- **Vehicle**

Candidate Keys	{ Vehicle_ID }
BCNF Check	Only determinant is Vehicle_ID (the PK / superkey). No other non-trivial FDs exist within this relation.

Verdict	IN BCNF. No decomposition needed.
----------------	-----------------------------------

- **Transaction**

Candidate Keys	{ Transaction_ID }
BCNF Check	Only determinant is Transaction_ID (the PK / superkey). No other non-trivial FDs exist within this relation.
Verdict	IN BCNF. No decomposition needed.

- **Monthly_Pass**

Candidate Keys	{ Pass_ID }
BCNF Check	Pass_ID is the only determinant and it is the primary key (superkey). No partial or transitive dependencies.
Verdict	IN BCNF. No decomposition needed.

- **Fastag_Account**

Candidate Keys	{ Fastag_ID }
BCNF Check	Fastag_ID is the superkey and is the only determinant. No violations.
Verdict	IN BCNF. No decomposition needed.

- **Violation**

Candidate Keys	{ Violation_ID }
BCNF Check	Violation_ID is the only determinant and it is the primary key. No partial or transitive dependencies.
Verdict	IN BCNF. No decomposition needed.

- **Transaction_Pass, Transaction_Fastag, Transaction_Rate**

Candidate Keys	{ Transaction_ID, Pass_ID } / { Transaction_ID, Fastag_ID } / { Transaction_ID, Rate_ID }
BCNF Check	Relations consisting entirely of key attributes have no non-trivial FDs, so BCNF holds trivially.
Verdict	IN BCNF. No decomposition needed.

➔ Here all relations are in BCNF so no decomposition required.

. PostgreSQL DDL Script

```
CREATE SCHEMA toll_management;
```

```
SET search_path TO toll_management;
```

-- Toll Plaza

```
CREATE TABLE Toll_Plaza (
    Plaza_ID PRIMARY KEY,
    Plaza_Name VARCHAR(100),
    Location VARCHAR(100),
    Highway_Name VARCHAR(100),
    Number_of_Lanes INT,
    Contact_Number VARCHAR(15)
);
```

-- Toll Booth

```
CREATE TABLE Toll_Booth (
    Booth_ID PRIMARY KEY,
    Booth_Number INT,
    Status VARCHAR(20),
    Plaza_ID INT REFERENCES Toll_Plaza(Plaza_ID)
);
```

-- Staff

```
CREATE TABLE Staff (  
    Staff_ID PRIMARY KEY,  
    Staff_Name VARCHAR(100),  
    Role VARCHAR(50),  
    Salary NUMERIC,  
    Booth_ID INT REFERENCES Toll_Booth(Booth_ID)  
);
```

-- Operates

```
CREATE TABLE Operates (  
    Staff_ID INT,  
    Booth_ID INT,  
    Shift_Start TIMESTAMP,  
    Shift_End TIMESTAMP,  
    PRIMARY KEY (Staff_ID, Booth_ID, Shift_Start),  
    FOREIGN KEY (Staff_ID) REFERENCES Staff(Staff_ID),  
    FOREIGN KEY (Booth_ID) REFERENCES Toll_Booth(Booth_ID)  
);
```

-- Vehicle

```
CREATE TABLE Vehicle (  
    Vehicle_ID PRIMARY KEY,  
    Vehicle_Number VARCHAR(20),  
    Vehicle_Type VARCHAR(50),  
    Owner_Name VARCHAR(100),  
    Owner_Contact_Number VARCHAR(15)  
);
```

-- Transaction

```
CREATE TABLE Transaction (  
    Transaction_ID PRIMARY KEY,  
    Vehicle_ID INT REFERENCES Vehicle(Vehicle_ID),  
    Booth_ID INT REFERENCES Toll_Booth(Booth_ID),  
    Transaction_Time TIMESTAMP,  
    Amount_Paid NUMERIC,  
    Payment_Mode VARCHAR(50)  
);
```

-- Monthly Pass

```
CREATE TABLE Monthly_Pass (  
    Pass_ID PRIMARY KEY,  
    Vehicle_ID INT REFERENCES Vehicle(Vehicle_ID),  
    Plaza_ID INT REFERENCES Toll_Plaza(Plaza_ID),  
    Expiry_Date DATE,  
    Pass_Fee NUMERIC  
);
```

-- FASTag Account

```
CREATE TABLE FASTag_Account (  
    Fastag_ID PRIMARY KEY,  
    Vehicle_ID INT REFERENCES Vehicle(Vehicle_ID),  
    Balance NUMERIC,  
    Bank_Name VARCHAR(100)  
);
```

-- Violation

```
CREATE TABLE Violation (  
    Violation_ID PRIMARY KEY,  
    Vehicle_ID INT REFERENCES Vehicle(Vehicle_ID),  
    Plaza_ID INT REFERENCES Toll_Plaza(Plaza_ID),
```

```
Violation_Type VARCHAR(100),  
Violation_Date DATE,  
Penalty_Amount NUMERIC  
);
```

-- Toll Rate

```
CREATE TABLE Toll_Rate (  
    Plaza_ID INT,  
    Vehicle_Type VARCHAR(50),  
    Toll_Amount NUMERIC,  
    PRIMARY KEY (Plaza_ID, Vehicle_Type),  
    FOREIGN KEY (Plaza_ID) REFERENCES Toll_Plaza(Plaza_ID)  
);
```

-- MANY-TO-MANY RELATION TABLE

-- 1. Staff ↔ Plaza

```
CREATE TABLE Staff_Plaza (  
    Staff_ID INT,  
    Plaza_ID INT,  
    Assigned_Date DATE,  
    PRIMARY KEY (Staff_ID, Plaza_ID),  
    FOREIGN KEY (Staff_ID) REFERENCES Staff(Staff_ID),  
    FOREIGN KEY (Plaza_ID) REFERENCES Toll_Plaza(Plaza_ID)  
);
```

-- 2. Transaction ↔ Monthly Pass

```
CREATE TABLE Transaction_Pass (  
    Transaction_ID INT,  
    Pass_ID INT,
```



```
PRIMARY KEY (Transaction_ID, Pass_ID),  
FOREIGN KEY (Transaction_ID) REFERENCES Transaction(Transaction_ID),  
FOREIGN KEY (Pass_ID) REFERENCES Monthly_Pass(Pass_ID)  
);
```

-- 3. Transaction ↔ FASTag

```
CREATE TABLE Transaction_FASTag (  
    Transaction_ID INT,  
    Fastag_ID INT,  
    PRIMARY KEY (Transaction_ID, Fastag_ID),  
    FOREIGN KEY (Transaction_ID) REFERENCES Transaction(Transaction_ID),  
    FOREIGN KEY (Fastag_ID) REFERENCES FASTag_Account(Fastag_ID)  
);
```

-- 4. Transaction ↔ Toll Rate

```
CREATE TABLE Transaction_Rate (  
    Transaction_ID INT,  
    Plaza_ID INT,  
    Vehicle_Type VARCHAR(50),  
    PRIMARY KEY (Transaction_ID, Plaza_ID, Vehicle_Type),  
    FOREIGN KEY (Transaction_ID) REFERENCES Transaction(Transaction_ID),  
    FOREIGN KEY (Plaza_ID, Vehicle_Type)  
        REFERENCES Toll_Rate(Plaza_ID, Vehicle_Type)  
);
```